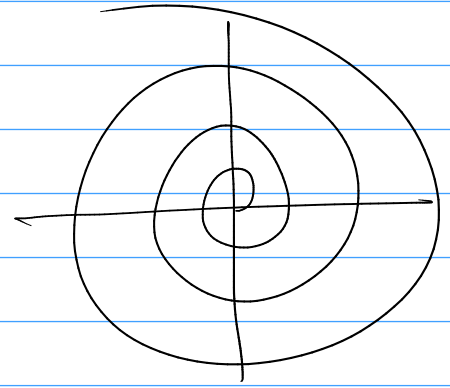
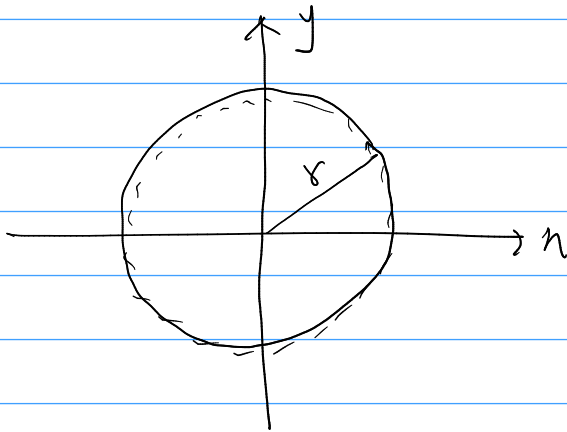


Circular Motion



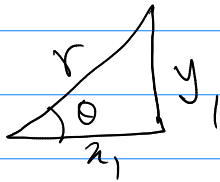
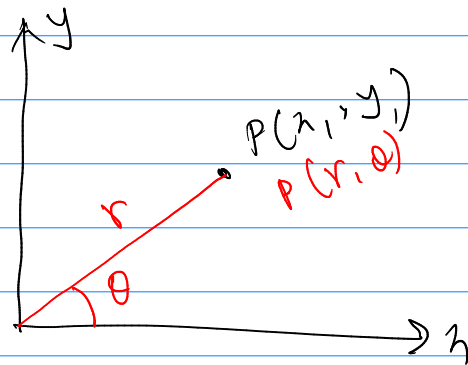
Polar coordinate system?

$$x_1 = r \cos \theta$$

$$y_1 = r \sin \theta$$

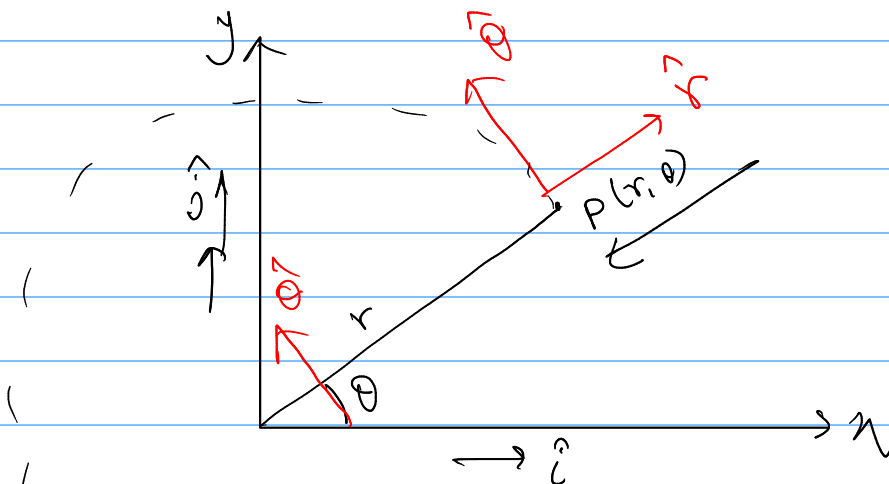
$$\vec{r} = x_1 \hat{i} + y_1 \hat{j}$$

$$\boxed{\vec{r} = r \cos \theta \hat{i} + r \sin \theta \hat{j}}$$



$$\vec{v} = v_x \hat{i} + v_y \hat{j}$$

$$\vec{v} = v_r \hat{r} + v_\theta \hat{\theta}$$



$$\hat{r} = \frac{\vec{r}}{|\vec{r}|}$$

$$\hat{r} = \frac{r \cos \theta \hat{i} + r \sin \theta \hat{j}}{r}$$

$$\boxed{\hat{r} = \cos \theta \hat{i} + \sin \theta \hat{j}}$$

$$\boxed{\hat{\theta} = -\sin \theta \hat{i} + \cos \theta \hat{j}}$$

$$\frac{d}{dt}(uv \dots w) = \frac{du}{dt}v + u \frac{dv}{dt} + \dots - uv \frac{dw}{dt}$$

Position:

$$\vec{r} = r \hat{r}$$

$$\vec{r} = x \hat{i}$$

velocity:

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r \hat{r})$$

$$\hat{r} = \cos\theta \hat{i} + \sin\theta \hat{j}$$

$$\Rightarrow \vec{v} = \frac{dr}{dt} \hat{r} + r \frac{d\hat{r}}{dt}$$

$$\frac{d\hat{r}}{dt} \cdot \frac{d\theta}{dt}$$

$$\Rightarrow \vec{v} = \frac{dr}{dt} \hat{r} + r \frac{d\hat{r}}{dt} \frac{d\theta}{dt}$$

$$\hat{r} = \cos\theta \hat{i} + \sin\theta \hat{j}$$

$$\Rightarrow \vec{v} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}$$

$$\frac{d\hat{r}}{dt} = -\sin\theta \hat{i} + \cos\theta \hat{j}$$

$$\Rightarrow \vec{v} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}$$

$$\frac{d\hat{r}}{dt} = \hat{\theta}$$

$$\vec{v} = v_x \hat{i} + v_y \hat{j}$$

$$\frac{dr}{dt} = \dot{r}$$

$$\frac{d\theta}{dt} = \dot{\theta}$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$\vec{a} = \frac{d}{dt}(\dot{r} \hat{r} + r \dot{\theta} \hat{\theta})$$

$$\vec{a} = (\ddot{r} - r \dot{\theta}^2) \hat{r} + (r \ddot{\theta} + 2\dot{r} \dot{\theta}) \hat{\theta}$$

Uniform circular motion

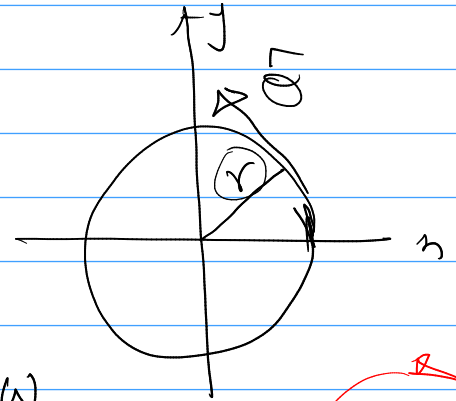
$$\vec{v} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}$$

$$\dot{r} = 0$$

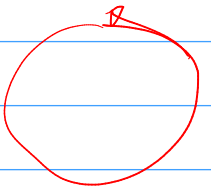
$$\vec{v} = r \dot{\theta} \hat{\theta}$$

$$\vec{v} = \omega r \hat{\theta}$$

$$\dot{\theta} = \frac{d\theta}{dt} = \omega$$



$$v = \omega r$$



$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt} (r \omega \hat{\theta})$$

$$\vec{v} = \omega \vec{r}$$

$$\vec{a} = r \omega \frac{d\hat{\theta}}{dt}$$

$$\hat{\theta} = -\sin\theta \hat{i} + \cos\theta \hat{j}$$

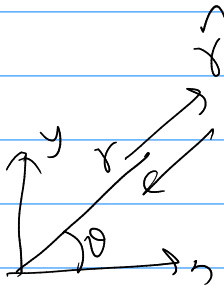
$$\vec{a} = r \omega \frac{d\hat{\theta}}{d\theta} \frac{d\theta}{dt}$$

$$\frac{d\hat{\theta}}{d\theta} = -(\cos\theta \hat{i} + \sin\theta \hat{j})$$

$$\vec{a} = r \omega (-\hat{r}) \omega$$

$$\frac{d\hat{\theta}}{d\theta} = -\hat{r}$$

$$\vec{a} = -\omega^2 r \hat{r}$$



$$\vec{a}_c = -\omega^2 r \hat{r}$$

$$v = \omega r$$

$$\vec{a}_c = -\frac{v^2}{r} \hat{r}$$

$$\vec{a}_c = -\frac{v^2}{r} \hat{r}$$

$$\vec{F}_c = m \vec{a} = -m \frac{v^2}{r} \hat{r}$$

$$v = \omega r \quad \omega = \frac{2\pi f}{T}$$

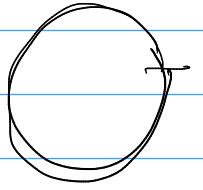
$$a_c = 398.5 \text{ m/s}^2$$

$$r = 10 \text{ cm}$$

$$600 \text{ rpm}$$

$$T = \frac{1}{10}$$

$$f = 0.1 \text{ s}^{-1}$$



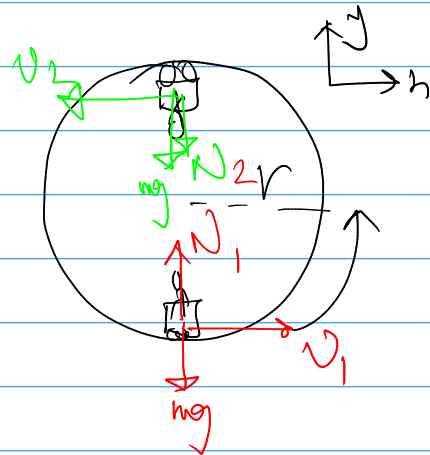
Roller coaster:

At bottom:

$$\sum F_y = ma$$

$$\Rightarrow N_1 - mg = \frac{mv_1^2}{r}$$

$$\Rightarrow N_1 = mg + \frac{mv_1^2}{r}$$



At top:

$$\sum F_y = ma$$

$$\Rightarrow -N_2 - mg = -\frac{mv_2^2}{r}$$

$$\Rightarrow N_2 + mg = \frac{mv_2^2}{r}$$

$$\Rightarrow N_2 = \frac{mv_2^2}{r} - mg$$

$$\frac{mv_2^2}{r} < mg$$

$$N_2 = -$$

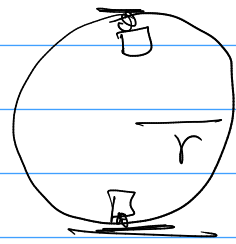
$$\frac{mv_2^2}{r} > mg$$

$$N = +$$

$$\frac{mv_2^2}{r} \geq mg \quad 4-9$$

$$\frac{mv_2^2}{r} = mg$$

$$\Rightarrow \boxed{v_2 = \sqrt{rg}}$$



$$E_b = E_t$$

$$\Rightarrow K_b + U_b = K_t + U_t$$

$$\Rightarrow \frac{1}{2}mv_1^2 + 0 = \frac{1}{2}mv_2^2 + mg(2r)$$

$$\Rightarrow v_1^2 = v_2^2 + 4gr$$

$$\Rightarrow v_1^2 = rg + 4gr$$

$$\Rightarrow \boxed{v_1 = \sqrt{5rg}}$$

Ferris wheel!

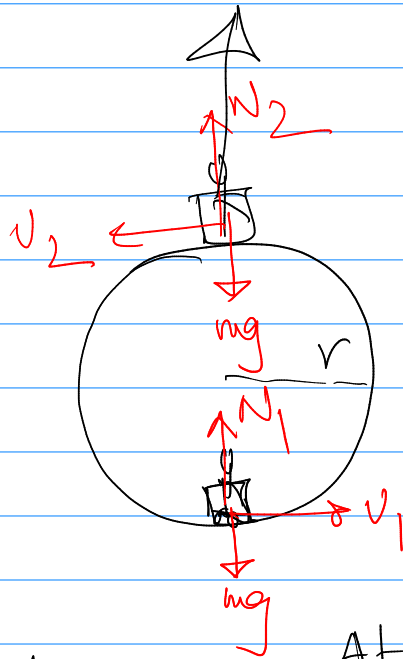
At bottom:

$$\sum F_y = ma$$

$$\Rightarrow N_1 - mg = \frac{mv_1^2}{r}$$

$$\Rightarrow \boxed{N_1 = mg + \frac{mv_1^2}{r}}$$

$$\frac{mv_2^2}{r} > mg$$

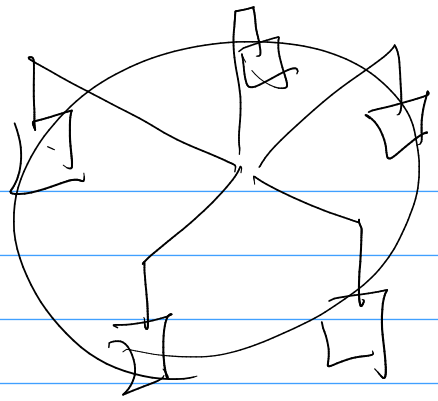


At top point:

$$\sum F_y = ma$$

$$N_2 - mg = -\frac{mv_2^2}{r}$$

$$\Rightarrow \boxed{N_2 = mg - \frac{mv_2^2}{r}}$$



$$y = y_0 + v_y t + \frac{1}{2} a_y t^2$$

$$v = \frac{v_h + v_y}{t}$$